

12.2 Fourier Series

INTRODUCTION We have just seen in the preceding section that if $\{\phi_0(x), \phi_1(x), \phi_2(x), \dots\}$ is a set of real-valued functions that is orthogonal on an interval $[a, b]$ and if f is a function defined on the same interval, then we can formally expand f in an orthogonal series $c_0\phi_0(x) + c_1\phi_1(x) + c_2\phi_2(x) + \dots$. In this section we shall expand functions in terms of a special orthogonal set of trigonometric functions.

Trigonometric Series In Problem 12 in Exercises 12.1, you were asked to show that the set of trigonometric functions

$$\left\{ 1, \cos \frac{\pi}{p} x, \cos \frac{2\pi}{p} x, \cos \frac{3\pi}{p} x, \dots, \sin \frac{\pi}{p} x, \sin \frac{2\pi}{p} x, \sin \frac{3\pi}{p} x, \dots \right\} \quad (1)$$

is orthogonal on the interval $[-p, p]$. This set will be of special importance later on in the solution of certain kinds of boundary-value problems involving linear partial differential equations. In those applications we will need to expand a function f defined on $[-p, p]$ in an orthogonal series consisting of the trigonometric functions in (1); that is,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi}{p} x + b_n \sin \frac{n\pi}{p} x \right). \quad (2)$$

The coefficients $a_0, a_1, a_2, \dots, b_1, b_2, \dots$, can be determined in exactly the same manner as in the general discussion of orthogonal series expansions on pages 674 and 675. Before proceeding, note that we have chosen to write the coefficient of 1 in the set (1) as $\frac{1}{2}a_0$ rather than a_0 ; this is for convenience only because the formula of a_n will then reduce to a_0 for $n = 0$.

This is why $\frac{1}{2}a_0$ is used instead of a_0 .

Now integrating both sides of (2) from $-p$ to p gives

$$\int_{-p}^p f(x) dx = \frac{a_0}{2} \int_{-p}^p dx + \sum_{n=1}^{\infty} \left(a_n \int_{-p}^p \cos \frac{n\pi}{p} x dx + b_n \int_{-p}^p \sin \frac{n\pi}{p} x dx \right). \quad (3)$$

Since $\cos(n\pi x/p)$ and $\sin(n\pi x/p)$, $n \geq 1$, are orthogonal to 1 on the interval, the right side of (3) reduces to a single term:

$$\int_{-p}^p f(x) dx = \frac{a_0}{2} \int_{-p}^p dx = \frac{a_0}{2} x \Big|_{-p}^p = pa_0.$$

Solving for a_0 yields

$$a_0 = \frac{1}{p} \int_{-p}^p f(x) dx. \quad (4)$$

Now we multiply (2) by $\cos(m\pi x/p)$ and integrate:

$$\begin{aligned} \int_{-p}^p f(x) \cos \frac{m\pi}{p} x dx &= \frac{a_0}{2} \int_{-p}^p \cos \frac{m\pi}{p} x dx \\ &+ \sum_{n=1}^{\infty} \left(a_n \int_{-p}^p \cos \frac{m\pi}{p} x \cos \frac{n\pi}{p} x dx + b_n \int_{-p}^p \cos \frac{m\pi}{p} x \sin \frac{n\pi}{p} x dx \right). \end{aligned} \quad (5)$$

By orthogonality we have

$$\int_{-p}^p \cos \frac{m\pi}{p} x dx = 0, \quad m > 0, \quad \int_{-p}^p \cos \frac{m\pi}{p} x \sin \frac{n\pi}{p} x dx = 0$$

and
$$\int_{-p}^p \cos \frac{m\pi}{p} x \cos \frac{n\pi}{p} x dx = \begin{cases} 0, & m \neq n \\ p, & m = n. \end{cases}$$

Thus (5) reduces to
$$\int_{-p}^p f(x) \cos \frac{n\pi}{p} x dx = a_n p,$$

and so
$$a_n = \frac{1}{p} \int_{-p}^p f(x) \cos \frac{n\pi}{p} x dx. \quad (6)$$

Finally, if we multiply (2) by $\sin(m\pi x/p)$, integrate, and make use of the results

$$\int_{-p}^p \sin \frac{m\pi}{p} x dx = 0, \quad m > 0, \quad \int_{-p}^p \sin \frac{m\pi}{p} x \cos \frac{n\pi}{p} x dx = 0$$

and
$$\int_{-p}^p \sin \frac{m\pi}{p} x \sin \frac{n\pi}{p} x dx = \begin{cases} 0, & m \neq n \\ p, & m = n, \end{cases}$$

we find that
$$b_n = \frac{1}{p} \int_{-p}^p f(x) \sin \frac{n\pi}{p} x dx. \quad (7)$$

The trigonometric series (2) with coefficients a_0 , a_n , and b_n defined by (4), (6), and (7), respectively, is said to be the **Fourier series** of the function f . The coefficients obtained from (4), (6), and (7) are referred to as **Fourier coefficients** of f .

In finding the coefficients a_0 , a_n , and b_n , we assumed that f was integrable on the interval and that (2), as well as the series obtained by multiplying (2) by $\cos(m\pi x/p)$, converged in such a manner as to permit term-by-term integration. Until (2) is shown to be convergent for a given function f , the equality sign is not to be taken in a strict or literal sense. Some texts use the symbol $\sim$ in place of $=$. In view of the fact that most functions in applications are of a type that guarantees convergence of the series, we shall use the equality symbol. We summarize the results:

Definition 12.2.1 Fourier Series

The **Fourier series** of a function f defined on the interval $(-p, p)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi}{p} x + b_n \sin \frac{n\pi}{p} x \right), \quad (8)$$

where

$$a_0 = \frac{1}{p} \int_{-p}^p f(x) dx \quad (9)$$

$$a_n = \frac{1}{p} \int_{-p}^p f(x) \cos \frac{n\pi}{p} x dx \quad (10)$$

$$b_n = \frac{1}{p} \int_{-p}^p f(x) \sin \frac{n\pi}{p} x dx. \quad (11)$$

EXAMPLE 1 Expansion in a Fourier Series

Expand
$$f(x) = \begin{cases} 0, & -\pi < x < 0 \\ \pi - x, & 0 \leq x < \pi \end{cases} \quad (12)$$

in a Fourier series.

SOLUTION The graph of f is given in **FIGURE 12.2.1**. With $p = \pi$ we have from (9) and (10) that

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \left[\int_{-\pi}^0 0 dx + \int_0^{\pi} (\pi - x) dx \right] = \frac{1}{\pi} \left[\pi x - \frac{x^2}{2} \right]_0^{\pi} = \frac{\pi}{2}$$

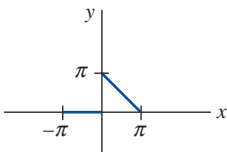


FIGURE 12.2.1 Function f in Example 1

$$\begin{aligned}
a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx = \frac{1}{\pi} \left[\int_{-\pi}^0 0 \, dx + \int_0^{\pi} (\pi - x) \cos nx \, dx \right] \\
&= \frac{1}{\pi} \left[(\pi - x) \frac{\sin nx}{n} \Big|_0^{\pi} + \frac{1}{n} \int_0^{\pi} \sin nx \, dx \right] \\
&= -\frac{1}{n\pi} \frac{\cos nx}{n} \Big|_0^{\pi} \\
&= \frac{-\cos n\pi + 1}{n^2\pi} \quad \leftarrow \cos n\pi = (-1)^n \\
&= \frac{1 - (-1)^n}{n^2\pi}.
\end{aligned}$$

In like manner we find from (11) that

$$b_n = \frac{1}{\pi} \int_0^{\pi} (\pi - x) \sin nx \, dx = \frac{1}{n}.$$

Note that

$$1 - (-1)^n = \begin{cases} 0, & n \text{ even} \\ 2, & n \text{ odd.} \end{cases}$$



Therefore
$$f(x) = \frac{\pi}{4} + \sum_{n=1}^{\infty} \left\{ \frac{1 - (-1)^n}{n^2\pi} \cos nx + \frac{1}{n} \sin nx \right\}. \quad (13) \equiv$$

Note that a_n defined by (10) reduces to a_0 given by (9) when we set $n = 0$. But as Example 1 shows, this may not be the case *after* the integral for a_n is evaluated.

Convergence of a Fourier Series The following theorem gives sufficient conditions for convergence of a Fourier series at a point.

Theorem 12.2.1 Conditions for Convergence

Let f and f' be piecewise continuous on the interval $[-p, p]$; that is, let f and f' be continuous except at a finite number of points in the interval and have only finite discontinuities at these points. Then for all x in the interval $(-p, p)$ the Fourier series of f converges to $f(x)$ at a point of continuity. At a point of discontinuity, the Fourier series converges to the average

$$\frac{f(x+) + f(x-)}{2},$$

where $f(x+)$ and $f(x-)$ denote the limit of f at x from the right and from the left, respectively.*

For a proof of this theorem you are referred to the classic text by Churchill and Brown.[†]

EXAMPLE 2 Convergence of a Point of Discontinuity

The function (12) in Example 1 satisfies the conditions of Theorem 12.2.1. Thus for every x in the interval $(-\pi, \pi)$, except at $x = 0$, the series (13) will converge to $f(x)$. At $x = 0$ the function is discontinuous, and so the series (13) will converge to

$$\frac{f(0+) + f(0-)}{2} = \frac{\pi + 0}{2} = \frac{\pi}{2}. \quad \equiv$$

* In other words, for x a point in the interval and $h > 0$,

$$f(x+) = \lim_{h \rightarrow 0} f(x + h), \quad f(x-) = \lim_{h \rightarrow 0} f(x - h).$$

[†] Ruel V. Churchill and James Ward Brown, *Fourier Series and Boundary Value Problems* (New York: McGraw-Hill, 2000).

We may assume that the given function f is periodic.

Periodic Extension Observe that each of the functions in the basic set (1) has a different fundamental period,* namely, $2p/n$, $n \geq 1$, but since a positive integer multiple of a period is also a period we see that all of the functions have in common the period $2p$ (verify). Hence the right-hand side of (2) is $2p$ -periodic; indeed, $2p$ is the fundamental period of the sum. We conclude that a Fourier series not only represents the function on the interval $(-p, p)$ but also gives the **periodic extension** of f outside this interval. We can now apply Theorem 12.2.1 to the periodic extension of f , or we may assume from the outset that the given function is periodic with period $T = 2p$; that is, $f(x + T) = f(x)$. When f is piecewise continuous and the right- and left-hand derivatives exist at $x = -p$ and $x = p$, respectively, then the series (8) converges to the average $[f(p-) + f(-p+)]/2$ at these endpoints and to this value extended periodically to $\pm 3p$, $\pm 5p$, $\pm 7p$, and so on. The Fourier series in (13) converges to the periodic extension of (12) on the entire x -axis. At 0 , $\pm 2\pi$, $\pm 4\pi$, ..., and at $\pm \pi$, $\pm 3\pi$, $\pm 5\pi$, ..., the series converges to the values

$$\frac{f(0+) + f(0-)}{2} = \frac{\pi}{2} \quad \text{and} \quad \frac{f(\pi-) + f(-\pi+)}{2} = 0,$$

respectively. The solid dots in **FIGURE 12.2.2** represent the value $\pi/2$.

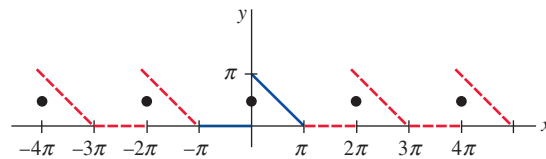


FIGURE 12.2.2 Periodic extension of the function f shown in Figure 12.2.1

Sequence of Partial Sums It is interesting to see how the sequence of partial sums $\{S_N(x)\}$ of a Fourier series approximates a function. For example, the first three partial sums of (13) are

$$S_1(x) = \frac{\pi}{4}, \quad S_2(x) = \frac{\pi}{4} + \frac{2}{\pi} \cos x + \sin x, \quad S_3(x) = \frac{\pi}{4} + \frac{2}{\pi} \cos x + \sin x + \frac{1}{2} \sin 2x.$$

In **FIGURE 12.2.3** we have used a CAS to graph the partial sums $S_5(x)$, $S_8(x)$, and $S_{15}(x)$ of (13) on the interval $(-\pi, \pi)$. Figure 12.2.3(d) shows the periodic extension using $S_{15}(x)$ on $(-4\pi, 4\pi)$.

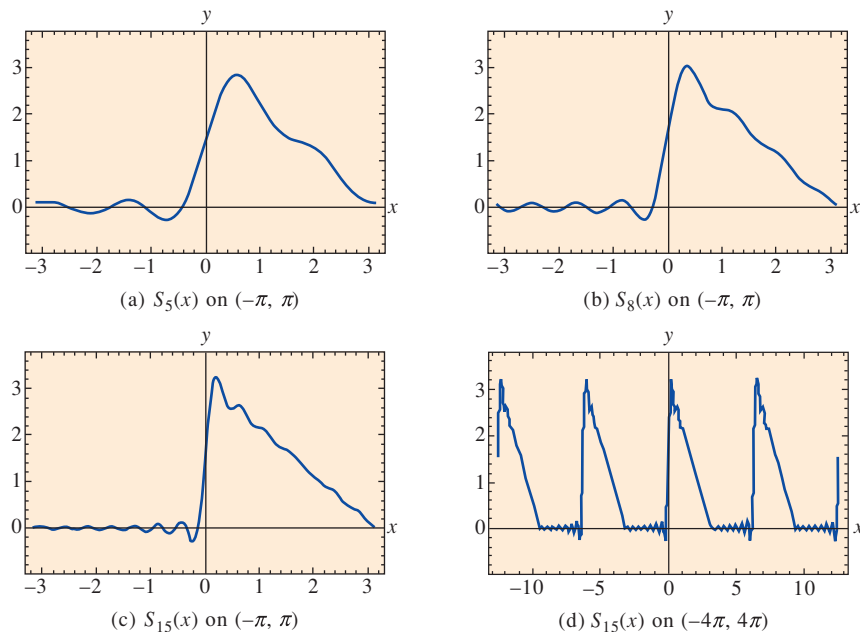


FIGURE 12.2.3 Partial sums of a Fourier series

* See Problems 21–26 in Exercises 12.1.